

Practice Set: Introduction to Statistical Methods

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Introduction

This document contains **45 fully solved problems**. The solutions are designed not just to give the answer, but to teach the underlying concept. We assume no prior steps are obvious and breakdown every calculation.

1 Module 1: Descriptive Statistics

Q1. Consider the following dataset representing the processing time (in ms) of a server:

$$\{12, 15, 14, 19, 18, 18, 17, 15, 16, 15, 14, 15, 16, 17, 18, 55\}$$

1. Determine the five-number summary (Min, Q1, Median, Q3, Max).
2. Determine potential outliers using the $1.5 \times IQR$ rule.
3. Comment on the skewness of the data.

Detailed Walkthrough

Part 1: The Five-Number Summary

Logic: To find quartiles and the median, data must first be ordered from least to greatest.

- **Raw Data:** $\{12, 15, 14, 19, 18, 18, 17, 15, 16, 15, 14, 15, 16, 17, 18, 55\}$
- **Step 1: Sorting.** Let's arrange these 16 numbers in ascending order:

$$12, 14, 14, 15, 15, 15, 15, 16, 16, 17, 17, 18, 18, 18, 19, 55$$

- **Step 2: Identify Min and Max.**

$$\text{Min} = 12, \quad \text{Max} = 55$$

- **Step 3: Find the Median (Q2).** The sample size is $n = 16$ (an even number). The median is the average of the two middle terms (the 8th and 9th terms).

$$8\text{th term} = 16, \quad 9\text{th term} = 16$$

$$\text{Median} = \frac{16 + 16}{2} = 16$$

- **Step 4: Find the First Quartile (Q1).** Q1 is the median of the *first half* of the data (the first 8 numbers): $\{12, 14, 14, 15, 15, 15, 15, 16\}$. Since there are 8 numbers (even), Q1 is the average of the 4th and 5th terms.

$$4\text{th} = 15, \quad 5\text{th} = 15 \implies Q1 = \frac{15 + 15}{2} = 15$$

- **Step 5: Find the Third Quartile (Q3).** Q3 is the median of the *second half* of the data (the last 8 numbers): {16, 17, 17, 18, 18, 18, 19, 55}. Average of the 4th and 5th terms of this subset (which are the 12th and 13th overall).

$$12\text{th} = 18, \quad 13\text{th} = 18 \implies Q3 = \frac{18 + 18}{2} = 18$$

Summary: Min=12, Q1=15, Median=16, Q3=18, Max=55.

Part 2: Outlier Detection

Logic: We construct a "fence" using the Interquartile Range (IQR). Anything outside the fence is an outlier.

- **Calculate IQR:** $IQR = Q3 - Q1 = 18 - 15 = 3$.
- **Calculate Upper Fence:** $Q3 + (1.5 \times IQR) = 18 + (1.5 \times 3) = 18 + 4.5 = 22.5$.
- **Calculate Lower Fence:** $Q1 - (1.5 \times IQR) = 15 - 4.5 = 10.5$.
- **Observation:** Are there values > 22.5 or < 10.5 ? Yes, the value **55** is greater than 22.5.

Conclusion: 55 is an outlier.

Part 3: Skewness Logic: Compare Mean vs Median.

- Median = 16.
- Mean = Sum/16 = 294/16 = 18.375.
- Since Mean (18.375) $>$ Median (16), the tail is pulled to the right by the outlier.

Conclusion: The distribution is **Positively Skewed** (Right-Skewed).

Q2. A factory has two machines. Machine A produces bolts with a mean length of 50mm and standard deviation 2mm. Machine B produces bolts with a mean length of 100mm and standard deviation 3mm. Which machine is more consistent?

Detailed Walkthrough

Concept: Standard Deviation (σ) measures absolute spread. However, when comparing datasets with significantly different means (μ), σ is misleading. We must use the **Coefficient of Variation (CV)**, which measures relative variability.

$$CV = \left(\frac{\sigma}{\mu} \right) \times 100\%$$

Step 1: Analyze Machine A

- $\mu_A = 50, \sigma_A = 2$.
- $CV_A = \frac{2}{50} \times 100\% = 0.04 \times 100\% = 4\%$.

Step 2: Analyze Machine B

- $\mu_B = 100, \sigma_B = 3.$
- $CV_B = \frac{3}{100} \times 100\% = 0.03 \times 100\% = 3\%.$

Comparison: A lower CV implies less variability relative to the size of the mean. Since $3\% < 4\%$, Machine B has less relative variation.

Final Answer: Machine B is more consistent.

Q3. The mean salary of 50 employees is 20 LPA. Later it was found that two salaries were wrongly recorded as 18 and 12 instead of 28 and 32. Find the correct mean.

Detailed Walkthrough

Concept: The Mean is $\frac{\text{Sum of Observations}}{\text{Total Count}}$. To fix the mean, we must fix the Sum first.

Step 1: Calculate the Incorrect Sum

$$\text{Sum}_{old} = n \times \bar{X}_{old}$$

Given $n = 50$ and $\bar{X}_{old} = 20$.

$$\text{Sum}_{old} = 50 \times 20 = 1000$$

Step 2: Perform the Correction We must remove the wrong values and add the correct values.

- Wrong values to subtract: 18, 12.
- Correct values to add: 28, 32.

$$\text{Sum}_{new} = 1000 - (18 + 12) + (28 + 32)$$

$$\text{Sum}_{new} = 1000 - 30 + 60$$

$$\text{Sum}_{new} = 1030$$

Step 3: Calculate New Mean

$$\bar{X}_{new} = \frac{\text{Sum}_{new}}{n} = \frac{1030}{50}$$

$$\bar{X}_{new} = \frac{103}{5} = 20.6$$

Final Answer: The correct mean is 20.6 LPA.

2 Module 2: Probability and Bayes' Theorem

Q4. Let A and B be two events. $P(A) = 0.4$, $P(B) = 0.5$, and $P(A \cap B) = 0.2$. 1. Find $P(A \cup B)$. 2. Find $P(A|B)$. 3. Are A and B independent?

Detailed Walkthrough

Part 1: Union of Events Using the Additive Law of Probability:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Substitute the values:

$$P(A \cup B) = 0.4 + 0.5 - 0.2 = 0.7$$

Interpretation: There is a 70% chance that either A or B (or both) happens.

Part 2: Conditional Probability Using the definition of Conditional Probability $P(A|B)$ (Probability of A given B has happened):

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Substitute the values:

$$P(A|B) = \frac{0.2}{0.5} = \frac{2}{5} = 0.4$$

Part 3: Independence Check *Concept:* Events are independent if the occurrence of one does not affect the probability of the other. Mathematically, $P(A \cap B) = P(A) \cdot P(B)$.

- Left Hand Side (Given): $P(A \cap B) = 0.2$
- Right Hand Side (Calculated): $P(A) \times P(B) = 0.4 \times 0.5 = 0.2$

Since LHS = RHS, the events are **Independent**.

Q5 (Important 2025 Pattern). A rare disease affects 0.1% of the population. A test has 99% sensitivity (True Positive Rate) and 98% specificity (True Negative Rate). If a person tests positive, what is the probability they actually have the disease?

Detailed Walkthrough**Step 1: Define Events**

- D : Person has the Disease.
- H : Person is Healthy (No Disease).
- $+ve$: Test result is Positive.
- $-ve$: Test result is Negative.

Step 2: Extract Probabilities

- **Prior:** $P(D) = 0.1\% = 0.001$.
- **Complement Prior:** $P(H) = 1 - P(D) = 0.999$.

- **Sensitivity:** $P(+ve|D) = 0.99$ (True Positive).
- **Specificity:** $P(-ve|H) = 0.98$ (True Negative).
- **False Positive:** $P(+ve|H) = 1 - P(-ve|H) = 1 - 0.98 = 0.02$.

Step 3: Bayes' Theorem Formula We want $P(D|+ve)$:

$$P(D|+ve) = \frac{P(+ve|D) \cdot P(D)}{P(+ve)}$$

Where $P(+ve)$ (Total Probability) is:

$$P(+ve) = P(+ve|D)P(D) + P(+ve|H)P(H)$$

Step 4: Calculation

- Numerator (True Positives): $0.99 \times 0.001 = 0.00099$.
- Denominator (All Positives):

$$\begin{aligned} & (0.99 \times 0.001) + (0.02 \times 0.999) \\ &= 0.00099 + 0.01998 = 0.02097 \end{aligned}$$

Step 5: Final Division

$$P(D|+ve) = \frac{0.00099}{0.02097} \approx 0.0472$$

Interpretation: Even with a positive test, there is only a **4.72%** chance you actually have the disease. This occurs because the disease is extremely rare (0.1%), so the False Positives overwhelm the True Positives.

Q6. In a factory, Machine A produces 60% of items, and Machine B produces 40%. Machine A has a defect rate of 2%, and Machine B has a defect rate of 3%. An item is chosen and found to be defective. What is the probability it came from Machine B?

Detailed Walkthrough

Step 1: Define Events

- A : Item from Machine A. $P(A) = 0.60$.
- B : Item from Machine B. $P(B) = 0.40$.
- D : Item is Defective.

Step 2: Conditional Probabilities (Likelihoods)

- $P(D|A) = 0.02$ (2% defect rate).
- $P(D|B) = 0.03$ (3% defect rate).

Step 3: Goal Find $P(B|D)$ using Bayes' Theorem.

$$P(B|D) = \frac{P(D|B) \cdot P(B)}{P(D)}$$

Step 4: Calculate Total Probability of Defect $P(D)$

$$P(D) = P(D|A)P(A) + P(D|B)P(B)$$

$$P(D) = (0.02 \times 0.60) + (0.03 \times 0.40)$$

$$P(D) = 0.012 + 0.012 = 0.024$$

Step 5: Calculate Posterior Probability

$$P(B|D) = \frac{0.012}{0.024} = \frac{1}{2} = 0.5$$

Final Answer: There is a **50%** chance the item came from Machine B.

Q7. You have three cards: one red on both sides (RR), one green on both sides (GG), and one red on one side and green on the other (RG). You pick a card blindly and see one red face. What is the probability the other side is also red?

Detailed Walkthrough

Step 1: Identify the Sample Space of Visible Faces We are picking a card and looking at one side. There are $3 \text{ cards} \times 2 \text{ sides} = 6 \text{ sides total}$.

- Card 1 (RR): Side 1A (Red), Side 1B (Red).
- Card 2 (GG): Side 2A (Green), Side 2B (Green).
- Card 3 (RG): Side 3A (Red), Side 3B (Green).

Step 2: Apply the Condition "See Red" We only care about outcomes where the visible face is Red.

- Outcome 1: Side 1A (Red) from Card RR.
- Outcome 2: Side 1B (Red) from Card RR.
- Outcome 3: Side 3A (Red) from Card RG.

Total possible outcomes = 3.

Step 3: Identify Favorable Outcomes We want the **other side** to be Red.

- If we see Side 1A, the other side is Side 1B (Red). (Success)
- If we see Side 1B, the other side is Side 1A (Red). (Success)

- If we see Side 3A, the other side is Side 3B (Green). (Failure)

Favorable outcomes = 2.

Step 4: Calculate Probability

$$P = \frac{\text{Favorable Outcomes}}{\text{Total Possible Outcomes}} = \frac{2}{3}$$

Final Answer: 2/3. (This is often counter-intuitive; many guess 1/2).

3 Module 3: Naive Bayes (Bag of Words)

Q8 (2025 Midsem Pattern). A system classifies messages as Spam or Ham. Training Data:

- Spam (2 msgs): "Win money now", "Win free cash"
- Ham (2 msgs): "Call me now", "Meeting call"

Test Message: "Win call" Using Naive Bayes with **Laplace Smoothing**, classify the message.

Detailed Walkthrough

Phase 1: Training (Building the Model)

Step 1: Vocabulary (V) List all unique words found in the training data: $V = \{\text{Win, money, now, free, cash, Call, me, Meeting}\}$. Size of Vocabulary $|V| = 8$.

Step 2: Prior Probabilities Total Docs = 4. Spam = 2. Ham = 2.

$$P(\text{Spam}) = \frac{2}{4} = 0.5, \quad P(\text{Ham}) = \frac{2}{4} = 0.5$$

Step 3: Word Counts per Class

- **Spam Class:** Words: "Win, money, now, Win, free, cash". Total words (N_{Spam}) = 6. Counts: Win=2, money=1, now=1, free=1, cash=1. (Others=0).
- **Ham Class:** Words: "Call, me, now, Meeting, call". (Assuming case-insensitive: Call=call). Total words (N_{Ham}) = 5. Counts: Call=2, me=1, now=1, Meeting=1. (Others=0).

Phase 2: Testing (Classifying "Win call")

We need to calculate two scores using Laplace Smoothing:

$$P(w|c) = \frac{\text{count}(w, c) + 1}{N_c + |V|}$$

Step 4: Calculate Conditional Probabilities For Spam ($N_{\text{Spam}} = 6, |V| = 8$): Denominator = $6 + 8 = 14$.

- $P(\text{"Win"}|Spam) = (2 + 1)/14 = 3/14$.
- $P(\text{"call"}|Spam) = (0 + 1)/14 = 1/14$.

For Ham ($N_{Ham} = 5, |V| = 8$): Denominator = $5 + 8 = 13$.

- $P(\text{"Win"}|Ham) = (0 + 1)/13 = 1/13$.
- $P(\text{"call"}|Ham) = (2 + 1)/13 = 3/13$.

Step 5: Calculate Posterior Scores Score = Prior $\times$ Likelihood(Word1) $\times$ Likelihood(Word2)

Spam Score:

$$0.5 \times \frac{3}{14} \times \frac{1}{14} = 0.5 \times \frac{3}{196} \approx 0.5 \times 0.0153 = \mathbf{0.00765}$$

Ham Score:

$$0.5 \times \frac{1}{13} \times \frac{3}{13} = 0.5 \times \frac{3}{169} \approx 0.5 \times 0.0177 = \mathbf{0.00887}$$

Conclusion: Since the Ham score (0.00887) is higher than the Spam score (0.00765), the message "Win call" is classified as **Ham**.

4 Module 4: Random Variables

Q9. A random variable X has the PMF: $P(X = x) = kx^2$ for $x = -1, 0, 1$ and 0 otherwise. 1. Find k . 2. Find $E[X]$. 3. Find $Var(X)$.

Detailed Walkthrough

Part 1: Finding Constant k The sum of all probabilities in a PMF must equal 1.

$$\sum P(X = x) = 1$$

$$P(-1) + P(0) + P(1) = 1$$

$$k(-1)^2 + k(0)^2 + k(1)^2 = 1$$

$$k(1) + 0 + k(1) = 1 \implies 2k = 1 \implies k = 0.5$$

So the probabilities are: $P(-1) = 0.5, P(0) = 0, P(1) = 0.5$.

Part 2: Expectation $E[X]$ Formula: $E[X] = \sum xP(x)$

$$E[X] = (-1)(0.5) + (0)(0) + (1)(0.5)$$

$$E[X] = -0.5 + 0 + 0.5 = 0$$

Part 3: Variance $Var(X)$ Formula: $Var(X) = E[X^2] - (E[X])^2$. First, calculate $E[X^2] = \sum x^2P(x)$:

$$E[X^2] = (-1)^2(0.5) + (0)^2(0) + (1)^2(0.5)$$

$$E[X^2] = (1)(0.5) + 0 + (1)(0.5) = 1$$

Now, calculate Variance:

$$\text{Var}(X) = 1 - (0)^2 = 1$$

Q10. A continuous random variable X has PDF $f(x) = c(4x - 2x^2)$ for $0 < x < 2$. 1. Find c . 2. Find $P(X > 1)$.

Detailed Walkthrough

Part 1: Finding c The total area under a PDF must equal 1.

$$\int_0^2 c(4x - 2x^2) dx = 1$$

Pull out c and integrate term by term:

$$c \left[\frac{4x^2}{2} - \frac{2x^3}{3} \right]_0^2 = 1$$

Simplify terms ($4/2 = 2$):

$$c \left[2x^2 - \frac{2x^3}{3} \right]_0^2 = 1$$

Substitute upper limit (2):

$$c \left[2(2)^2 - \frac{2(2)^3}{3} \right] = c \left[8 - \frac{16}{3} \right]$$

Find common denominator: $8 = 24/3$.

$$c \left[\frac{24}{3} - \frac{16}{3} \right] = c \left[\frac{8}{3} \right] = 1$$

$$c = \frac{3}{8}$$

Part 2: Probability $P(X > 1)$ We integrate the PDF from 1 to 2.

$$P(X > 1) = \int_1^2 \frac{3}{8}(4x - 2x^2) dx$$

$$= \frac{3}{8} \left[2x^2 - \frac{2x^3}{3} \right]_1^2$$

Calculate value at Upper Limit (2): We already know this is $8/3$ from Part 1. Calculate value at Lower Limit (1):

$$2(1)^2 - \frac{2(1)^3}{3} = 2 - \frac{2}{3} = \frac{4}{3}$$

Subtract Lower from Upper:

$$\begin{aligned}\text{Integral} &= \frac{3}{8} \left(\frac{8}{3} - \frac{4}{3} \right) = \frac{3}{8} \left(\frac{4}{3} \right) \\ &= \frac{12}{24} = 0.5\end{aligned}$$

Q11. Given $E[X] = 2$ and $\text{Var}(X) = 3$. Find the mean and variance of $Y = 3X - 5$.

Detailed Walkthrough

Step 1: Properties of Expectation The expectation operator $E[\cdot]$ is linear.

$$E[aX + b] = aE[X] + b$$

Substitute $a = 3, b = -5, E[X] = 2$:

$$E[Y] = 3(2) - 5 = 6 - 5 = 1$$

Step 2: Properties of Variance Variance is affected by scaling (a^2) but not by shifting (b).

$$\text{Var}(aX + b) = a^2 \text{Var}(X)$$

Substitute $a = 3, \text{Var}(X) = 3$:

$$\text{Var}(Y) = 3^2 \times 3 = 9 \times 3 = 27$$

Final Answer: Mean = 1, Variance = 27.

5 Module 5: Standard Distributions

Q12. A biased coin has $P(H) = 0.2$. It is tossed 5 times. What is the probability of getting exactly 2 heads?

Detailed Walkthrough

Step 1: Identify Distribution

- Fixed number of trials: $n = 5$.
- Two outcomes (Head/Tail): Binary.
- Constant probability: $p = 0.2$.
- Independent trials.

This is a **Binomial Distribution** $B(n, p)$.

Step 2: Formula

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

We need $P(X = 2)$.

Step 3: Calculation

$$P(X = 2) = \binom{5}{2} (0.2)^2 (0.8)^3$$

Calculate the Combination $\binom{5}{2}$:

$$\binom{5}{2} = \frac{5!}{2!(5-2)!} = \frac{5 \times 4}{2 \times 1} = 10$$

Calculate powers:

$$(0.2)^2 = 0.04$$

$$(0.8)^3 = 0.512$$

Multiply:

$$P(X = 2) = 10 \times 0.04 \times 0.512$$

$$P(X = 2) = 0.4 \times 0.512 = 0.2048$$

Q13. A website receives 3 visits per minute on average. Find the probability of receiving exactly 4 visits in the next minute using Poisson distribution.

Detailed Walkthrough

Step 1: Identify Parameters We are given a rate $\lambda = 3$ visits/minute. We want to find the probability of $k = 4$ events in that interval.

Step 2: Formula

$$P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}$$

Step 3: Substitution

$$P(X = 4) = \frac{e^{-3} 3^4}{4!}$$

Values:

- $e^{-3} \approx 0.049787$
- $3^4 = 81$
- $4! = 4 \times 3 \times 2 \times 1 = 24$

Step 4: Calculation

$$P(X = 4) = \frac{0.049787 \times 81}{24}$$

$$P(X = 4) = \frac{4.0327}{24} \approx 0.168$$

Q14 (Poisson approx to Binomial). A machine produces items where 1% are defective. In a batch of 200 items, find the probability of exactly 3 defectives using: a) Binomial formula (setup only), b) Poisson approximation (calculate).

Detailed Walkthrough

Parameters: $n = 200$ (Sample size). $p = 0.01$ (Probability of success/defect). $k = 3$ (Target successes).

Part A: Binomial Setup Formula: $P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$.

$$P(X = 3) = \binom{200}{3} (0.01)^3 (0.99)^{197}$$

(Calculating 0.99^{197} by hand is difficult, hence we use approximation).

Part B: Poisson Approximation Condition: n is large (> 20) and p is small (< 0.05), and np is moderate. Calculate Lambda λ :

$$\lambda = n \times p = 200 \times 0.01 = 2$$

Now use Poisson formula with $\lambda = 2, k = 3$:

$$\begin{aligned} P(X = 3) &= \frac{e^{-2} 2^3}{3!} \\ &= \frac{0.1353 \times 8}{6} \\ &= \frac{1.0824}{6} \approx 0.1804 \end{aligned}$$

Q15. Salaries are normally distributed with $\mu = 50k$ and $\sigma = 10k$. Find the probability that a person earns between 45k and 60k. (Assume Z-values: $\phi(0.5) = 0.6915, \phi(1) = 0.8413$).

Detailed Walkthrough

Step 1: Standardize the Variable Convert raw scores (X) to Standard Z-scores using $Z = \frac{X - \mu}{\sigma}$.

- Lower Bound $X = 45$:

$$Z_1 = \frac{45 - 50}{10} = \frac{-5}{10} = -0.5$$

- Upper Bound $X = 60$:

$$Z_2 = \frac{60 - 50}{10} = \frac{10}{10} = 1.0$$

Step 2: Express Probability in terms of Z We need $P(45 < X < 60)$, which becomes $P(-0.5 < Z < 1.0)$. This is the area under the curve between -0.5 and 1.0.

$$Area = \phi(1.0) - \phi(-0.5)$$

Step 3: Handle Negative Z The Normal table (ϕ) usually gives area to the left of positive numbers. By symmetry: $\phi(-0.5) = 1 - \phi(0.5)$. Given $\phi(0.5) = 0.6915$,

$$\phi(-0.5) = 1 - 0.6915 = 0.3085$$

Step 4: Final Subtraction Given $\phi(1.0) = 0.8413$.

$$P = 0.8413 - 0.3085 = 0.5328$$

6 Module 6: Joint Distributions

Q16. The joint PMF of X and Y is given by $P(x, y) = c(x + y)$ for $x = 1, 2$ and $y = 1, 2$. 1. Find c . 2. Find the marginal distribution of X. 3. Find $P(Y = 1|X = 2)$.

Detailed Walkthrough

Part 1: Find Constant c The sum of probabilities over the entire grid must be 1. Grid Points (x, y) : $(1, 1), (1, 2), (2, 1), (2, 2)$. Calculate $P(x, y)$ for each:

- $P(1, 1) = c(1 + 1) = 2c$
- $P(1, 2) = c(1 + 2) = 3c$
- $P(2, 1) = c(2 + 1) = 3c$
- $P(2, 2) = c(2 + 2) = 4c$

$$\text{Sum} = 2c + 3c + 3c + 4c = 12c.$$

$$12c = 1 \implies c = \frac{1}{12}$$

Part 2: Marginal Distribution of X ($P_X(x)$) To find probability of a specific x , sum over all possible y values.

- For $X = 1$: $\text{sum } P(1, 1) + P(1, 2) = 2c + 3c = 5c = 5/12$.
- For $X = 2$: $\text{sum } P(2, 1) + P(2, 2) = 3c + 4c = 7c = 7/12$.

Check: $5/12 + 7/12 = 12/12 = 1$. (Correct).

Part 3: Conditional Probability

$$P(Y = 1|X = 2) = \frac{P(X = 2, Y = 1)}{P(X = 2)}$$

Numerator ($x = 2, y = 1$): $3c = 3/12$. Denominator ($x = 2$ marginal): $7c = 7/12$.

$$\text{Result} = \frac{3/12}{7/12} = \frac{3}{7}$$

7 Detailed Solutions for Questions 17–45

Q17. If $X \sim N(0, 1)$, calculate $P(|Z| > 1.5)$. (Given $\phi(1.5) = 0.9332$).

Detailed Walkthrough

Step 1: Understand Absolute Value Inequality The condition $|Z| > 1.5$ means the value is extremely positive OR extremely negative.

$$P(|Z| > 1.5) = P(Z > 1.5) + P(Z < -1.5)$$

Step 2: Use Symmetry The Standard Normal curve is symmetric about 0. The area in the right tail ($Z > 1.5$) is identical to the area in the left tail ($Z < -1.5$). So, Total Probability $= 2 \times P(Z > 1.5)$.

Step 3: Calculate One Tail The table value $\phi(1.5) = 0.9332$ represents the area to the left ($Z < 1.5$). The area to the right is the complement:

$$P(Z > 1.5) = 1 - \phi(1.5) = 1 - 0.9332 = 0.0668$$

Step 4: Final Calculation

$$P(|Z| > 1.5) = 2 \times 0.0668 = 0.1336$$

Q18. A data scientist observes that the number of errors in code follows a distribution where Mean = 5 and Variance = 4. Is this a Poisson distribution?

Detailed Walkthrough

Step 1: Recall Poisson Properties The Poisson distribution is defined by a single parameter λ (lambda). A unique and strict property of Poisson is:

$$\text{Mean}(\mu) = \text{Variance}(\sigma^2) = \lambda$$

Step 2: Check the Data

- Observed Mean = 5
- Observed Variance = 4

Step 3: Conclusion Since $5 \neq 4$, the condition is violated. The variance is less than the mean (Under-dispersion). **Answer:** No, this is not a Poisson distribution. It might be a Binomial distribution.

Q19. Given a sample size $n = 100$, sample mean $\bar{x} = 50$, standard deviation $s = 5$. Find the 95% confidence interval for the mean. ($Z_{0.025} = 1.96$).

Detailed Walkthrough

Step 1: Formula for Confidence Interval (CI) Since $n \geq 30$, we use the Z-statistic approximation.

$$CI = \bar{x} \pm \left(Z_{\alpha/2} \times \frac{s}{\sqrt{n}} \right)$$

The term $\frac{s}{\sqrt{n}}$ is the Standard Error (SE).

Step 2: Calculate Standard Error

$$SE = \frac{5}{\sqrt{100}} = \frac{5}{10} = 0.5$$

Step 3: Calculate Margin of Error (ME)

$$ME = Z \times SE = 1.96 \times 0.5 = 0.98$$

Step 4: Construct Interval

$$CI = 50 \pm 0.98$$

Lower Limit: $50 - 0.98 = 49.02$. Upper Limit: $50 + 0.98 = 50.98$. **Answer:** The 95% CI is $[49.02, 50.98]$.

Q20. Two dice are rolled. Let X be the sum. Find $P(X \geq 10)$.

Detailed Walkthrough

Step 1: Total Sample Space Two dice are rolled. Each has 6 faces. Total outcomes $N = 6 \times 6 = 36$.

Step 2: Identify Favorable Outcomes We need the sum S to be 10, 11, or 12.

- Sum = 10: (4, 6), (5, 5), (6, 4). (3 outcomes)
- Sum = 11: (5, 6), (6, 5). (2 outcomes)
- Sum = 12: (6, 6). (1 outcome)

Total favorable outcomes = $3 + 2 + 1 = 6$.

Step 3: Probability Calculation

$$P(X \geq 10) = \frac{\text{Favorable}}{\text{Total}} = \frac{6}{36}$$

Simplifying fraction:

$$P(X \geq 10) = \frac{1}{6} \approx 0.167$$

Q21. The joint PDF of X and Y is $f(x, y) = 4xy$ for $0 < x < 1, 0 < y < 1$. Are X and Y independent?

Detailed Walkthrough

Concept: Two random variables are independent if and only if their joint PDF factors into the product of their marginal PDFs:

$$f(x, y) = f_X(x) \cdot f_Y(y)$$

Step 1: Find Marginal PDF of X ($f_X(x)$) Integrate joint PDF w.r.t y (range 0 to 1):

$$\begin{aligned} f_X(x) &= \int_0^1 4xy \, dy = 4x \left[\frac{y^2}{2} \right]_0^1 \\ &= 4x \left(\frac{1}{2} - 0 \right) = 2x \end{aligned}$$

Step 2: Find Marginal PDF of Y ($f_Y(y)$) Integrate joint PDF w.r.t x (range 0 to 1):

$$\begin{aligned} f_Y(y) &= \int_0^1 4xy \, dx = 4y \left[\frac{x^2}{2} \right]_0^1 \\ &= 4y \left(\frac{1}{2} - 0 \right) = 2y \end{aligned}$$

Step 3: Verification Multiply marginals:

$$f_X(x) \cdot f_Y(y) = (2x) \cdot (2y) = 4xy$$

Since this equals the original $f(x, y)$, they are **Independent**.

Q22. Given data: 2, 4, 4, 4, 5, 5, 7, 9. Calculate the sample variance.

Detailed Walkthrough

Step 1: Calculate Mean ($\bar{x}$) Sum = $2 + 4 + 4 + 4 + 5 + 5 + 7 + 9 = 40$. Count $n = 8$. Mean $\bar{x} = 40/8 = 5$.

Step 2: Calculate Squared Deviations ($(x_i - \bar{x})^2$)

- 2: $(2 - 5)^2 = (-3)^2 = 9$
- 4: $(4 - 5)^2 = (-1)^2 = 1$ (occurs 3 times) $\rightarrow 3$
- 5: $(5 - 5)^2 = 0$ (occurs 2 times) $\rightarrow 0$
- 7: $(7 - 5)^2 = 2^2 = 4$
- 9: $(9 - 5)^2 = 4^2 = 16$

Sum of Squares (SS) = $9 + 1 + 1 + 1 + 0 + 0 + 4 + 16 = 32$.

Step 3: Sample Variance Formula For sample variance, we divide by $n - 1$ (Bessel's

correction).

$$s^2 = \frac{SS}{n-1} = \frac{32}{8-1} = \frac{32}{7} \approx 4.57$$

Q23. A box contains 3 Red and 2 Blue balls. Two balls are drawn **without replacement**. Find the probability that the second ball is Red given the first was Blue.

Detailed Walkthrough

Concept: Conditional probability changes the sample space. "Given the first was Blue" means we assume the event B_1 has already happened and changed the box state.

Step 1: Initial State Box: 3 Red, 2 Blue. Total = 5.

Step 2: Event B_1 (First is Blue) We remove 1 Blue ball. New State: 3 Red, 1 Blue. Total = 4.

Step 3: Event R_2 (Second is Red) We are drawing from the New State. Number of Red balls = 3. Total balls = 4.

$$P(R_2|B_1) = \frac{3}{4} = 0.75$$

Q24. Using the same box (3R, 2B), let X be the number of Red balls drawn in 2 draws. Find the probability distribution of X.

Detailed Walkthrough

Experiment: Draw 2 balls from 5 without replacement. Total ways to pick 2 balls: $\binom{5}{2} = 10$. Random Variable X: Count of Red balls (Possible values: 0, 1, 2).

Case X=0 (0 Red, 2 Blue): We must choose 0 Red from 3 AND 2 Blue from 2.

$$\binom{3}{0} \times \binom{2}{2} = 1 \times 1 = 1$$

$$P(X = 0) = 1/10.$$

Case X=1 (1 Red, 1 Blue): Choose 1 Red from 3 AND 1 Blue from 2.

$$\binom{3}{1} \times \binom{2}{1} = 3 \times 2 = 6$$

$$P(X = 1) = 6/10.$$

Case X=2 (2 Red, 0 Blue): Choose 2 Red from 3 AND 0 Blue from 2.

$$\binom{3}{2} \times \binom{2}{0} = 3 \times 1 = 3$$

$$P(X = 2) = 3/10.$$

Check: $1 + 6 + 3 = 10$. Sum of probabilities is 1.

Q25. $X \sim N(100, 15)$. Find the Z-score for $x = 130$. Is this an outlier?

Detailed Walkthrough

Step 1: Calculate Z-score Formula: $Z = \frac{x - \mu}{\sigma}$.

$$Z = \frac{130 - 100}{15} = \frac{30}{15} = 2.0$$

Step 2: Outlier Criteria In statistics, particularly normally distributed data:

- $Z > 3$: Extreme outlier.
- $Z > 2$: Significant, but usually not labeled an outlier unless using strict criteria.

Common rule: Outliers are beyond 3 standard deviations. **Answer:** $Z = 2$. It is not an outlier (it lies within 95% of data, not the extreme 0.3%).

Q26. In a Naive Bayes problem, if a word "Crypto" never appears in the "Ham" training set, what problem occurs and how do we fix it?

Detailed Walkthrough

The Problem: Zero Frequency Naive Bayes calculates the probability of a message belonging to a class by multiplying probabilities of individual words. If "Crypto" has count 0 in Ham, then $P(\text{"Crypto"}|\text{Ham}) = 0$. The entire product becomes:

$$P(\text{Ham}|\text{msg}) \propto P(\text{Ham}) \times \dots \times 0 \times \dots = 0$$

This forces the prediction to 0, ignoring all other valid words in the message.

The Solution: Laplace Smoothing (Additive Smoothing) We add a dummy count of 1 to every word frequency and normalize by adding the Vocabulary size ($|V|$) to the denominator.

$$P_{\text{smoothed}}(w|c) = \frac{\text{count}(w, c) + 1}{\text{count}(\text{total_in_c}) + |V|}$$

This ensures no probability is ever exactly zero.

Q27. Find the median of the continuous random variable with PDF $f(x) = 2x$ for $0 < x < 1$.

Detailed Walkthrough

Concept: The Median m divides the probability area exactly in half. Area from lower limit (0) to m must equal 0.5.

Calculation:

$$\int_0^m f(x) dx = 0.5$$

$$\int_0^m 2x dx = 0.5$$

$$[x^2]_0^m = 0.5$$

$$m^2 - 0 = 0.5$$

$$m = \sqrt{0.5}$$

$$m \approx 0.707$$

Since $0 < 0.707 < 1$, the median is within the valid range.

Q28. Given $P(A) = 0.3$, $P(B) = 0.4$. If A and B are mutually exclusive, find $P(A \cup B)$.

Detailed Walkthrough

Concept: "Mutually Exclusive" (or Disjoint) means the events cannot happen at the same time. Therefore, the intersection $P(A \cap B) = 0$.

Calculation:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A \cup B) = 0.3 + 0.4 - 0 = 0.7$$

Q29. If A and B are independent in Q28, find $P(A \cup B)$.

Detailed Walkthrough

Concept: "Independent" means the intersection is the product of individual probabilities.

$$P(A \cap B) = P(A) \times P(B)$$

$$P(A \cap B) = 0.3 \times 0.4 = 0.12$$

Calculation:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A \cup B) = 0.3 + 0.4 - 0.12$$

$$P(A \cup B) = 0.7 - 0.12 = 0.58$$

Q30. Compute $E[X]$ for a fair die roll.

Detailed Walkthrough

Setup: X takes values $\{1, 2, 3, 4, 5, 6\}$. Since the die is fair, $P(x) = 1/6$ for all x .

Calculation:

$$E[X] = \sum xP(x)$$

$$E[X] = 1\left(\frac{1}{6}\right) + 2\left(\frac{1}{6}\right) + 3\left(\frac{1}{6}\right) + 4\left(\frac{1}{6}\right) + 5\left(\frac{1}{6}\right) + 6\left(\frac{1}{6}\right)$$

$$E[X] = \frac{1 + 2 + 3 + 4 + 5 + 6}{6}$$

$$E[X] = \frac{21}{6} = 3.5$$

Q31. Compute $E[X^2]$ for a fair die roll.

Detailed Walkthrough

Calculation:

$$E[X^2] = \sum x^2P(x)$$

$$E[X^2] = \frac{1^2 + 2^2 + 3^2 + 4^2 + 5^2 + 6^2}{6}$$

Square the values: 1, 4, 9, 16, 25, 36.

$$E[X^2] = \frac{1 + 4 + 9 + 16 + 25 + 36}{6}$$

$$E[X^2] = \frac{91}{6} \approx 15.167$$

Q32. Using Q30 and Q31, find the Variance of a die roll.

Detailed Walkthrough

Formula:

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

From previous questions: $E[X^2] = 91/6$ $E[X] = 3.5 = 7/2$.

Calculation:

$$\text{Var}(X) = \frac{91}{6} - \left(\frac{7}{2}\right)^2$$

$$\text{Var}(X) = \frac{91}{6} - \frac{49}{4}$$

Find common denominator (12):

$$\frac{91 \times 2}{12} - \frac{49 \times 3}{12} = \frac{182 - 147}{12}$$

$$\text{Var}(X) = \frac{35}{12} \approx 2.916$$

Q33. A bag has 3 coins. C_1 is fair ($P(H) = 0.5$), C_2 is two-headed ($P(H) = 1.0$), C_3 is biased ($P(H) = 0.75$). A coin is picked at random and flipped. It shows Heads. What is the probability it is the two-headed coin (C_2)?

Detailed Walkthrough

Step 1: Priors Since we pick blindly from 3 coins: $P(C_1) = P(C_2) = P(C_3) = 1/3$.

Step 2: Likelihoods (Probability of Evidence given Class) Evidence $E = \text{Heads}$.

- $P(E|C_1) = 0.5$
- $P(E|C_2) = 1.0$
- $P(E|C_3) = 0.75$

Step 3: Total Probability of Evidence $P(E)$

$$\begin{aligned} P(E) &= \sum P(E|C_i)P(C_i) \\ P(E) &= \frac{1}{3}(0.5) + \frac{1}{3}(1.0) + \frac{1}{3}(0.75) \\ P(E) &= \frac{1}{3}(0.5 + 1.0 + 0.75) = \frac{2.25}{3} = 0.75 \end{aligned}$$

Step 4: Bayes' Theorem

$$\begin{aligned} P(C_2|E) &= \frac{P(E|C_2)P(C_2)}{P(E)} \\ P(C_2|E) &= \frac{1.0 \times (1/3)}{0.75} \\ P(C_2|E) &= \frac{0.3333}{0.75} = \frac{1/3}{3/4} = \frac{4}{9} \approx 0.44 \end{aligned}$$

Q34. Define a Bernoulli Trial.

Detailed Walkthrough

Definition: A Bernoulli trial is the simplest random experiment with the following specific properties:

1. **Binary Outcome:** There are exactly two possible outcomes, conventionally labeled "Success" (1) and "Failure" (0).
2. **Constant Probability:** The probability of success, denoted by p , is fixed and does not change.

Example: Flipping a single coin once.

Q35. What is the relationship between Bernoulli and Binomial distributions?

Detailed Walkthrough

Concept: The Binomial Distribution is essentially the **sum** of multiple Bernoulli trials.

- A Bernoulli trial is a single experiment ($n = 1$).
- A Binomial distribution ($B(n, p)$) models the number of successes in n independent, identical Bernoulli trials.

If $X_1, X_2, \dots, X_n$ are independent Bernoulli variables with parameter p , then $Y = \sum X_i$ follows a Binomial distribution.

Q36. If covariance $Cov(X, Y) = 0$, are X and Y independent?

Detailed Walkthrough

Answer: No, not necessarily. **Explanation:**

- **Independence $\implies$ Zero Covariance:** If variables are independent, they have no relationship, so covariance is 0.
- **Zero Covariance $\implies$ Independence:** Covariance only measures **linear** relationships. Variables can have a strong non-linear relationship (e.g., a circle or parabola) and still have zero covariance.

Counter-example: Let X be uniform on $[-1, 1]$ and $Y = X^2$. They are clearly dependent, but $Cov(X, Y) = 0$.

Q37. Given $X \sim Poi(4)$. Find $P(X \geq 2)$.

Detailed Walkthrough

Strategy: Calculating sum for $2, 3, 4, \dots, \infty$ is impossible. Use the Complement Rule.

$$P(X \geq 2) = 1 - P(X < 2)$$

$$P(X < 2) = P(X = 0) + P(X = 1)$$

Calculation: Formula: $P(x) = \frac{e^{-4} 4^x}{x!}$.

- $P(0) = e^{-4} \frac{4^0}{0!} = e^{-4}(1) \approx 0.0183$.

- $P(1) = e^{-4} \frac{4^1}{1!} = 4e^{-4} \approx 0.0732$.

Sum of complements = $0.0183 + 0.0732 = 0.0915$.

Final Answer:

$$P(X \geq 2) = 1 - 0.0915 = 0.9085$$

Q38. The joint PDF is $f(x, y) = k$ for $0 < x < 2, 0 < y < x$. Find k .

Detailed Walkthrough

Step 1: Visualize the Region The region is bounded by $x = 0, x = 2, y = 0$, and the line $y = x$. This forms a triangle with vertices $(0, 0), (2, 0), (2, 2)$.

Step 2: Calculate Area of Region Area of Triangle = $0.5 \times \text{base} \times \text{height}$. Base (on x-axis) = 2. Height (at $x=2, y=x$ implies $y=2$) = 2. Area = $0.5 \times 2 \times 2 = 2$.

Step 3: Solve for k For a uniform joint PDF, Volume = Area $\times$ Height(k) = 1.

$$2 \times k = 1$$

$$k = 0.5$$

Q39. For Q38, find the marginal PDF of X .

Detailed Walkthrough

Formula: Marginal $f_X(x) = \int f(x, y) dy$. We integrate out y . The limits for y are given as 0 to x .

Calculation:

$$f_X(x) = \int_0^x k dy$$

Since $k = 0.5$:

$$f_X(x) = \int_0^x 0.5 dy = 0.5[y]_0^x$$

$$f_X(x) = 0.5(x - 0) = 0.5x$$

Range: This is valid for $0 < x < 2$.

Q40. A student scores marks: 10, 20, 30, 40, 500. Which measure of central tendency is best?

Detailed Walkthrough

Analysis: The data set is $\{10, 20, 30, 40, 500\}$.

- **Mean:** $(10 + 20 + 30 + 40 + 500)/5 = 600/5 = 120$. Notice that 120 is higher than 4 out

of 5 observations. The value 500 is an extreme outlier.

- **Median:** The middle value is 30.

Conclusion: The **Median** is the best measure here because it is "robust" (resistant) to outliers. It represents the "typical" student score better than the inflated mean.

Q41. X is a continuous variable uniform on $[10, 20]$. Find $P(X < 12)$.

Detailed Walkthrough

Step 1: Identify PDF For Uniform $[a, b]$, PDF is constant $\frac{1}{b-a}$. Here $a = 10, b = 20$. Height of PDF = $\frac{1}{20-10} = \frac{1}{10} = 0.1$.

Step 2: Calculate Probability $P(X < 12)$ is the area of the rectangle from $x = 10$ to $x = 12$. Width = $12 - 10 = 2$. Height = 0.1 . Area = Width $\times$ Height = $2 \times 0.1 = 0.2$.

Final Answer: 0.2 (or 20%).

Q42. Can the Variance of a random variable be negative?

Detailed Walkthrough

Answer: No, never. **Mathematical Proof:** Variance is defined as $Var(X) = E[(X - \mu)^2]$. The term $(X - \mu)^2$ squares a real number, so the result is always non-negative (≥ 0). The Expectation (average) of non-negative values must be non-negative. Therefore, Variance ≥ 0 .

Q43. Calculate the correlation coefficient given $Cov(X, Y) = 3, Var(X) = 4, Var(Y) = 9$.

Detailed Walkthrough

Formula: Correlation Coefficient ρ is the covariance normalized by the standard deviations.

$$\rho_{X,Y} = \frac{Cov(X,Y)}{\sigma_X \sigma_Y}$$

Step 1: Get Standard Deviations $\sigma_X = \sqrt{Var(X)} = \sqrt{4} = 2$. $\sigma_Y = \sqrt{Var(Y)} = \sqrt{9} = 3$.

Step 2: Substitute

$$\rho = \frac{3}{2 \times 3} = \frac{3}{6} = 0.5$$

Interpretation: There is a moderate positive linear relationship.

Q44. A test has 5 questions (True/False). A student guesses randomly. What is the probability of getting at least 4 correct?

Detailed Walkthrough

Setup: Binomial Distribution. $n = 5$ trials. Probability of success (guessing correct T/F) $p = 0.5$. We want $P(X \geq 4) = P(X = 4) + P(X = 5)$.

Calculation:

$$P(X = 4) = \binom{5}{4} (0.5)^4 (0.5)^1 = 5 \times (0.5)^5$$

$$P(X = 5) = \binom{5}{5} (0.5)^5 (0.5)^0 = 1 \times (0.5)^5$$

$$\begin{aligned} \text{Total} &= (5 + 1) \times (0.5)^5 = 6 \times \frac{1}{32} \\ &= \frac{6}{32} = \frac{3}{16} = 0.1875 \end{aligned}$$

Q45. In hypothesis testing, if $p\text{-value} < 0.05$, what do we do?

Detailed Walkthrough

Explanation:

- The **p-value** measures the probability of seeing data this extreme if the Null Hypothesis (H_0) were true.
- The threshold 0.05 (alpha α) is the significance level.

Decision Rule: If $p < \alpha$, the event is "statistically significant" (too rare to happen by luck). Therefore, we **Reject the Null Hypothesis** (H_0) and accept the Alternative Hypothesis (H_1).

8 Module 7: Advanced Distributions & Hypothesis Testing Guide

Hypothesis Testing Framework Guide

Before solving the following problems, understand these key rules for framing hypotheses:

- **One-sided vs. Two-sided:**

- **Two-sided:** The question asks if groups are *different*, *changed*, or *not equal*. Direction doesn't matter.
 - * $H_0 : \mu = 50$, $H_1 : \mu \neq 50$.
- **One-sided:** The question asks if a group is *greater*, *less*, *improved*, or *decreased*.
 - * Right-tail: $H_0 : \mu \leq 50$, $H_1 : \mu > 50$ (e.g., "Is the new speed faster?").
 - * Left-tail: $H_0 : \mu \geq 50$, $H_1 : \mu < 50$ (e.g., "Has the defect rate dropped?").

- **Choosing the Right Test:**

- **Z-Test:** Use for means when Population SD (σ) is **Known** OR Sample size $n \geq 30$.
- **T-Test:** Use for means when Population SD (σ) is **Unknown** AND Sample size $n < 30$.
- **F-Test:** Use when comparing two **Variances** or Standard Deviations.
- **Chi-Square Test:** Use for **Categorical** data (counts/frequencies) or checking Independence.

Detailed Problems (Q46–Q59)

Q46 (Binomial). A basketball player has an 80% success rate for free throws. He takes 10 shots. Find the probability he makes exactly 7 shots.

Detailed Walkthrough

Step 1: Identify Parameters $n = 10$ trials. $p = 0.8$ (Success). $k = 7$ (Desired successes).

Step 2: Formula

$$P(X = 7) = \binom{10}{7} (0.8)^7 (0.2)^3$$

Use symmetry $\binom{10}{7} = \binom{10}{3} = \frac{10 \times 9 \times 8}{3 \times 2 \times 1} = 120$.

Step 3: Calculation

$$P(X = 7) = 120 \times (0.2097) \times (0.008)$$

$$P(X = 7) \approx 120 \times 0.001677 \approx 0.201$$

Answer: Approximately 20.1%.

Q47 (Binomial Cumulative). A fair coin is tossed 4 times. Find the probability of getting **at most** 2 heads.

Detailed Walkthrough

Concept: "At most 2" means $X \leq 2$, which includes $X = 0, X = 1, X = 2$.

Step 1: Individual Probabilities (Binomial $n = 4, p = 0.5$)

$$P(X = 0) = \binom{4}{0} (0.5)^4 = 1 \times \frac{1}{16} = \frac{1}{16}$$

$$P(X = 1) = \binom{4}{1} (0.5)^4 = 4 \times \frac{1}{16} = \frac{4}{16}$$

$$P(X = 2) = \binom{4}{2} (0.5)^4 = 6 \times \frac{1}{16} = \frac{6}{16}$$

Step 2: Summation

$$P(X \leq 2) = \frac{1}{16} + \frac{4}{16} + \frac{6}{16} = \frac{11}{16}$$

Answer: 0.6875.

Q48 (Poisson). A call center receives 4 calls per hour. What is the probability of receiving 0 calls in the next 30 minutes?

Detailed Walkthrough

Step 1: Adjust Lambda The rate is $\lambda = 4$ calls per 60 mins. For a 30 minute interval, the new rate $\lambda_{new} = \frac{4}{2} = 2$.

Step 2: Calculate

$$P(X = 0) = \frac{e^{-\lambda} \lambda^0}{0!}$$

$$P(X = 0) = \frac{e^{-2} \cdot 1}{1} = e^{-2}$$

Answer: ≈ 0.1353 .

Q49 (Poisson). Using the same rate ($\lambda = 2$ for 30 mins), find the probability of receiving at least 2 calls.

Detailed Walkthrough

Logic: Use Complement. $P(X \geq 2) = 1 - [P(0) + P(1)]$.

Step 1: Calculate P(1)

$$P(1) = \frac{e^{-2} 2^1}{1!} = 2e^{-2} \approx 2(0.1353) = 0.2706$$

Step 2: Sum and Subtract $P(0) = 0.1353$ (from Q48). Sum = $0.1353 + 0.2706 = 0.4059$.

$$P(X \geq 2) = 1 - 0.4059 = 0.5941$$

Q50 (Normal - One Sided Area). Scores are normally distributed with $\mu = 500, \sigma = 100$. A student needs to score in the top 10% to qualify. What is the minimum cutoff score? (Given $Z \approx 1.28$ for Area 0.90).

Detailed Walkthrough

Concept: Top 10% means the area to the LEFT is 90% (0.90). We look up the Z-table for probability 0.90, which corresponds to $Z = 1.28$.

Step 1: Inverse Z Formula

$$X = \mu + (Z \times \sigma)$$

$$X = 500 + (1.28 \times 100)$$

$$X = 500 + 128 = 628$$

Answer: The student must score at least 628.

Q51 (Normal - Symmetry). In a standard normal distribution, find k such that $P(-k < Z < k) = 0.95$.

Detailed Walkthrough

Concept: This is the logic behind a 95% Confidence Interval. If the middle area is 0.95, the remaining tails total $1 - 0.95 = 0.05$. Since it is symmetric, each tail has area 0.025. We need the Z-score where the area to the **left** is $0.95 + 0.025 = 0.975$. From standard Z-tables, Z for 0.975 is 1.96. **Answer:** $k = 1.96$.

Q52 (Z-Test Two-Sided). A manufacturer claims a bolt length is 10cm. SD is known $\sigma = 0.2$. A sample of 36 bolts has mean $\bar{x} = 10.1$. Test at $\alpha = 0.05$ if the mean is **different**.

Detailed Walkthrough

Step 1: Hypotheses $H_0 : \mu = 10$. $H_1 : \mu \neq 10$ (Two-sided).

Step 2: Calculate Z-statistic Since σ is known and $n = 36 \geq 30$:

$$Z_{calc} = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{10.1 - 10}{0.2/6}$$

$$Z_{calc} = \frac{0.1}{0.0333} = 3.0$$

Step 3: Conclusion Critical Z for $\alpha = 0.05$ (two-sided) is ± 1.96 . Since $|3.0| > 1.96$, we fall in the rejection region. **Result:** Reject H_0 . The mean length is significantly different.

Q53 (Z-Test One-Sided). Using the data from Q52, test if the mean length is **greater** than 10cm.

Detailed Walkthrough

Step 1: Hypotheses $H_0 : \mu \leq 10$. $H_1 : \mu > 10$ (One-sided, Right tail).

Step 2: Z-statistic $Z_{calc} = 3.0$ (Same calculation as above).

Step 3: Conclusion Critical Z for $\alpha = 0.05$ (one-sided) is $+1.645$. Since $3.0 > 1.645$, we reject H_0 . **Result:** The mean is statistically greater than 10cm.

Q54 (T-Test Two-Sided). A sample of $n = 16$ batteries has mean life 20 hours and sample SD $s = 2$. Test if the mean is different from 21 hours. ($\alpha = 0.05, t_{crit} = 2.131$).

Detailed Walkthrough

Choice of Test: $n < 30$ and population σ is Unknown. Use T-test.

Step 1: Hypotheses $H_0 : \mu = 21$. $H_1 : \mu \neq 21$.

Step 2: Calculate t-statistic

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{20 - 21}{2/\sqrt{16}}$$

$$t = \frac{-1}{2/4} = \frac{-1}{0.5} = -2.0$$

Step 3: Conclusion Degrees of Freedom $df = n - 1 = 15$. Critical value (Two-tailed, $\alpha = 0.05$) is given as 2.131. We compare $|t_{calc}|$ vs t_{crit} . $|-2.0| = 2.0$. Since $2.0 < 2.131$, we do NOT reject H_0 . **Result:** Not enough evidence to say the mean is different from 21.

Q55 (T-Test One-Sided). A teacher claims the average score is > 70 . Sample $n = 9$, $\bar{x} = 74$, $s = 6$. Test this claim. ($\alpha = 0.05$, $t_{crit, 1-tail, df=8} = 1.86$).

Detailed Walkthrough

Step 1: Hypotheses $H_0 : \mu \leq 70$. $H_1 : \mu > 70$ (Claim is "Greater").

Step 2: Calculate t

$$t = \frac{74 - 70}{6/\sqrt{9}} = \frac{4}{6/3} = \frac{4}{2} = 2.0$$

Step 3: Conclusion $t_{calc}(2.0) > t_{crit}(1.86)$. We fall in the rejection region. **Result:** Reject H_0 . The claim that the average is > 70 is supported.

Q56 (F-Test Setup). You want to compare the precision of two instruments. Instrument A: $n = 10$, $s^2 = 25$. Instrument B: $n = 10$, $s^2 = 10$. Calculate the F-statistic.

Detailed Walkthrough

Concept: F-test compares variances. We always place the larger variance in the numerator for the standard calculation.

Calculation:

$$F = \frac{s_1^2}{s_2^2}$$

Since $25 > 10$:

$$F = \frac{25}{10} = 2.5$$

Note: To finish the test, we would compare 2.5 against the F-table critical value for $df_1 = 9$, $df_2 = 9$.

Q57 (F-Test Critical Region). If you are testing if Variance A $\neq$ Variance B ($\alpha = 0.10$), is this one-sided or two-sided?

Detailed Walkthrough

Answer: Two-sided. **Explanation:** The hypothesis is $\sigma_A^2 \neq \sigma_B^2$. We care if A is significantly larger OR smaller than B. The alpha is split (0.05 in each tail). However, in manual F-test calculations, we typically arrange the ratio so $F > 1$ and only check the right tail of the F-distribution against $\alpha/2$.

Q58 (Chi-Square Goodness of Fit). A die is rolled 60 times. Observed counts: {1: 5, 2: 15, 3: 10, 4: 10, 5: 10, 6: 10}. Is the die fair?

Detailed Walkthrough

Step 1: Expected Counts If fair, $P(i) = 1/6$. Total $n = 60$. Expected (E) for each face = $60 \times (1/6) = 10$.

Step 2: Chi-Square Formula

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

Step 3: Calculation

- Face 1: $(5 - 10)^2/10 = 25/10 = 2.5$
- Face 2: $(15 - 10)^2/10 = 25/10 = 2.5$
- Faces 3,4,5,6: $(10 - 10)^2/10 = 0$

Total $\chi^2 = 2.5 + 2.5 + 0 + 0 + 0 + 0 = 5.0$.

Step 4: Degrees of Freedom $df = k - 1 = 6 - 1 = 5$. (Typically, critical value for $df = 5, \alpha = 0.05$ is 11.07. Since $5.0 < 11.07$, we do not reject fairness).

Q59 (Chi-Square Independence). A survey asks Gender (M/F) and Preference (Tea/Coffee). Data: [M, Tea: 20], [M, Coffee: 30], [F, Tea: 30], [F, Coffee: 20]. Calculate the expected value for the cell [Male, Tea].

Detailed Walkthrough

Step 1: Contingency Table Totals

	Tea	Coffee	Total
Male	20	30	50
Female	30	20	50
Total	50	50	100

Step 2: Formula for Expected Value

$$E_{row,col} = \frac{(\text{Row Total}) \times (\text{Col Total})}{\text{Grand Total}}$$

Step 3: Calculation for Male/Tea Row Total (Male) = 50. Col Total (Tea) = 50. Grand Total = 100.

$$E = \frac{50 \times 50}{100} = \frac{2500}{100} = 25$$

Interpretation: If Gender and Preference were independent, we would expect 25 Males to like Tea. We observed 20.